How to Read an Engineering Research Paper

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Reading research papers effectively is challenging. These papers are written in a very condensed style beca
and the intended audience, which is assumed to already know the area well. Moreover, the reasons for writi
different than the reasons the paper has been assigned, meaning you have to work harder to find the conten
in. Finally, your time is very limited, so you may not have time to read every word of the paper or read it seve
the nuances. For all these reasons, reading a research paper can require a special approach.

To develop an effective reading style for research papers, it can help to know two things: what you should ge
where that information is located in the paper. First, I'll describe how a typical research paper is put together

Despite a paper's condensed form, it is likely repetitive. The introduction will state not only the motivations b
outline the solution. Often this may be all the expert requires from the paper. The body of the paper states th
problem in detail, and should also describe a detailed evaluation of the solution in terms of arguments or an
paper will conclude with a recap, including a discussion of the primary contributions. A paper will also discus
degree. Because of the repetition in these papers at different levels of detail and from different perspectives
read the paper ``out of order'' or to skip certain sections. More on this below.

The questions you want to have answered by reading a paper are the following:

1. **What are the *motivations* for this work?** For a research paper, there is an expectation that a probler
no one else has published in the literature. This problem intrinsically has two parts. The first is often ur
people problem. The people problem is the benefits that are desired in the world at large; for example
life, such as saved time or increased safety. The second part is the **technical problem**, which is: *why*
problem have a trivial solution? There is also an implication that previous solutions to the problem are
previous solutions and why are they inadequate? Finally, the motivation and statement of the problem
research question, the question that the paper sets out to answer. This might be more focused than t
outset. Oftentimes, one or more of these elements are not explicitly stated, making your job more diffic

2. **What is the proposed *solution*?** This is also called the **hypothesis** or **idea**. This is the proposed ans
question. There should also be an answer to the question *why is it believed that this solution will work,*
previous solutions? There should also be a discussion about *how the solution is achieved (designed a*
least achievable.

3. **What is the work's *evaluation* of the proposed solution?** An idea alone is usually not adequate for paper. This is the concrete engagement of the research question. What argument, implementation, an the case for the value of the ideas? What benefits or problems are identified?

4. **What is your analysis of the identified problem, idea and evaluation?** Is this a good idea? What fl the work? What are the most interesting points made? What are the most controversial ideas or points practical implications, you also want to ask: *Is this really going to work, who would want it, what it will t when might it become a reality?*

5. **What are the *contributions*?** The contributions in a paper may be many and varied. Beyond the insig question, a few additional possibilities include: ideas, software, experimental techniques, or an area su

6. **What are *future directions* for this research?** Not only what future directions do the authors identify, come up with while reading the paper? Sometimes these may be identified as shortcomings or other c work.

7. **What questions are you left with?** What questions would you like to raise in an open discussion of th find confusing or difficult to understand? By taking the time to list several, you will be forced to think m work.

8. **What is your take-away message from this paper?** Sum up the main implication of the paper from y useful for very quick review and refreshing your memory. It also forces you to try to identify the essenc

As you read or skim a paper, you should actively attempt to answer the above questions. Presumably, the in motivation. The introduction and conclusion may discuss the solutions and evaluation at a high level. Future concluding part of the paper. The details of the solution and the evaluation should be in the body of the pape productive to try to answer each question in turn, writing your answer down. I recommend that you keep a n you read, or mark-up the papers themselves. You could use my standard two-page form that you can fill out practice, you are not done reading a paper until you can answer all the questions.

Also, you should be aware of the context of the paper in relation to the other papers in the class. Often a pa generalization, new direction, or contradiction to earlier papers.

If you find that filling out this form doesn't work for you, you can try writing a 250 word abstract of the paper- at the front of the paper, but *your* abstract, capturing the above five issues from your perspective. I often find abstract because it develops the logical connections between the above five issues.

Taking time to writing down *questions* you have about the paper will often surface thoughts that were not init the paper was vague on key issues, or ignored issues that you think are important. If you come to class with prepared to counter or preempt my own questions.

Reading a book is somewhat different. Although you want to answer the above questions for a book, it may given the amount of detail in each chapter. You may want to fill out the above questions on a chapter-by-cha produce a summary form for the entire book when you have finished reading it. However, each chapter will h that may make certain questions irrelevant. Also, a book is often not oriented towards explaining the solutior However, engineering books are invariably oriented towards problem solving of one kind or another.

I have a habit of writing on papers directly, less with books simply because they cost so much. A well-annota weight in gold, as it not only contains the content of the paper, but your assessment of its value to you.

Advice on note taking. Although I have provided a form that can be filled out, I actually advocate annotatin paper is a rich canvas on which to layer your thoughts. Here is how I suggest approaching the reading and r

- Highlight important comments as you go. Using a highlighter, as opposed to underlining, can really hel out" at you when you return to review the paper later.
- Mark the important paragraphs of the paper according to motivation/problem, idea/solution, their evalu
- On the front of the paper, write down the take-away message.
- On the front of the paper, or near the end, write down your key questions. Other questions may be writ read.
- Try to answer the questions for yourself, as best you can. Use Google or other sources as appropriate

Until you have been able to complete the above process, it is likely that you have not yet thought critically en second pass over the paper is sometimes required to have it all come together for you. To help you further s note-taking activities, you might want to follow this rubric, using it as a kind of check list.